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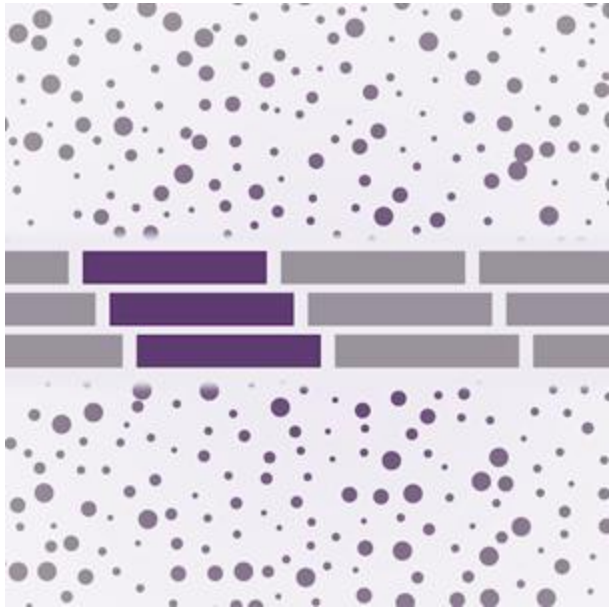
Canadian Bioinformatics Workshops

Module #1b: Sequencing Technology Applications

Dr. Erin Pleasance

CBW High-Throughput Genomics

November 4-6, 2026



CANADA'S MICHAEL SMITH GENOME SCIENCES CENTRE

Learning Objectives

- Considerations in analysing large genomic datasets
- Data sharing in genomics
- Overview of applications of sequencing technologies
- Considerations in choosing your approach

Genomics produces a lot of data … this brings complexities

- Data storage size
- Data sharing
- Statistical considerations

Data size

- One Illumina NovaSeqX 25B flowcell can create 10 Tb (10,000,000,000,000 bases) in one run
- Files are large – eg. one human genome bam/cram can be easily 50-300 GB
- Storage costs are significant
- Data processing can require significant compute
- Decisions about local vs. cloud storage/processing

Some common genomics databases

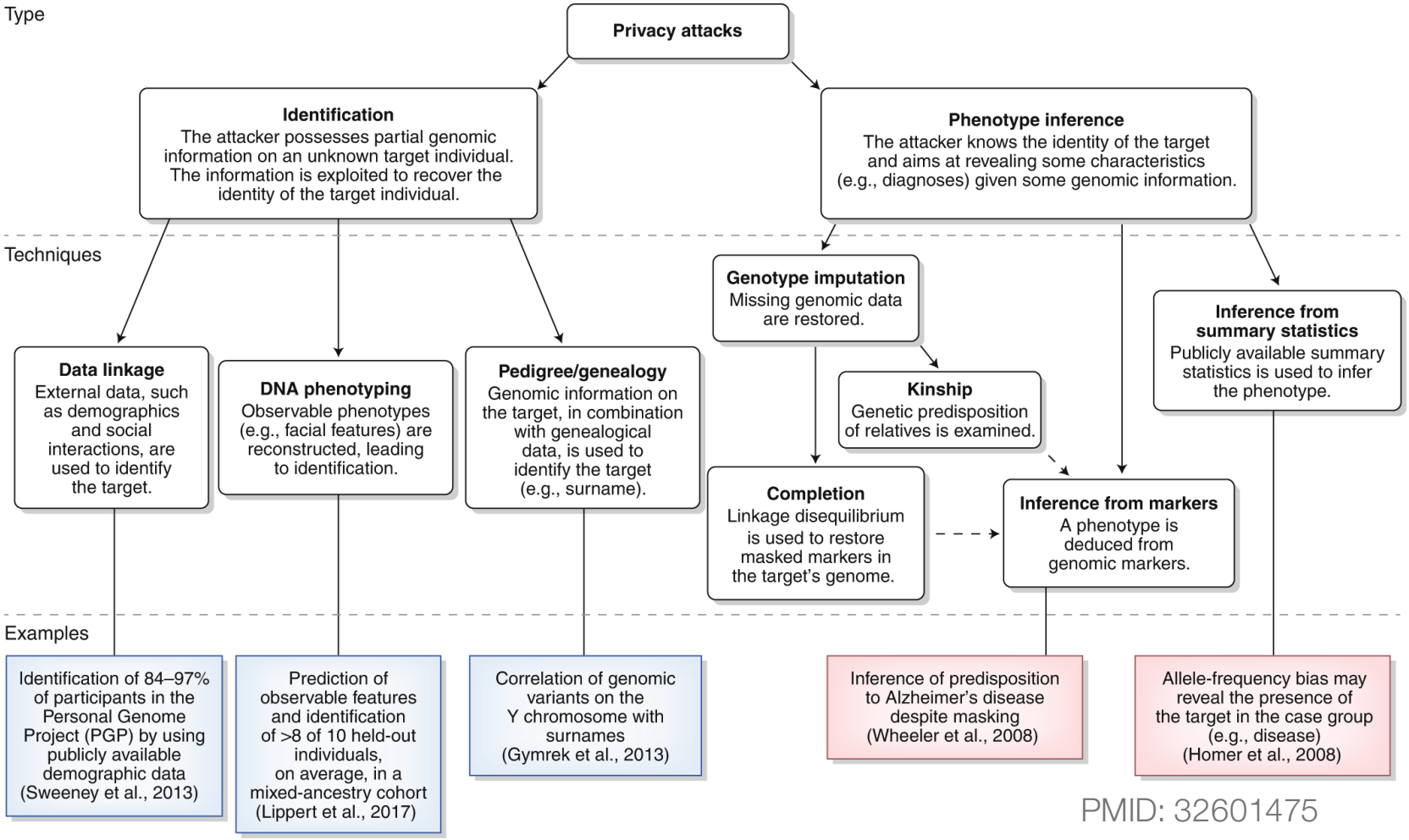
- SRA (NCBI): Publicly available large sequencing datasets
- EGA, dbGAP: Individual sequencing datasets with metadata
- GEO: Other genomics data eg. expression data
- dbSNP, DGV, ClinVar: Databases of variation

Data sharing in genomics

- Data sharing is central to genomics
- Findable, Accessible, Interoperable, and Reusable (FAIR) principles
  - Findable: Data are in a searchable database, with a unique ID, and metadata
  - Accessible: Data and metadata can be accessed in a standard way, with authorization as appropriate
  - Interoperable: Data and metadata are in standard format using standard ontologies
  - Reusable: Clear licensing, clear metadata, meet community standards
  - See https://www.nature.com/articles/sdata201618
- Required by many journals for publication
- Code sharing is also commonly required

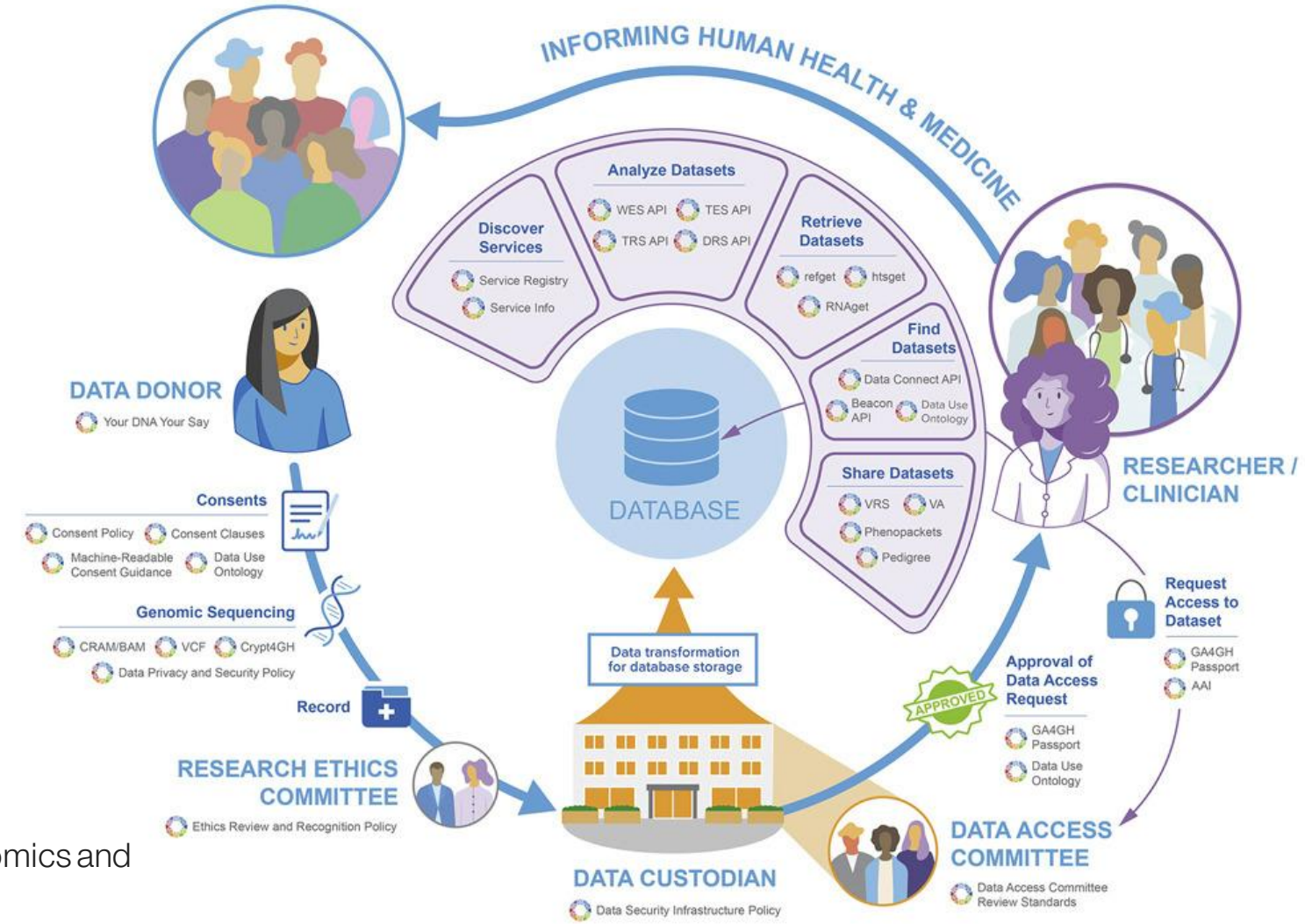
Data sharing and privacy

- Genomic data can be inherently identifiable
- Manage through anonymization and access control



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GA4GH Standards in the Data Life Cycle

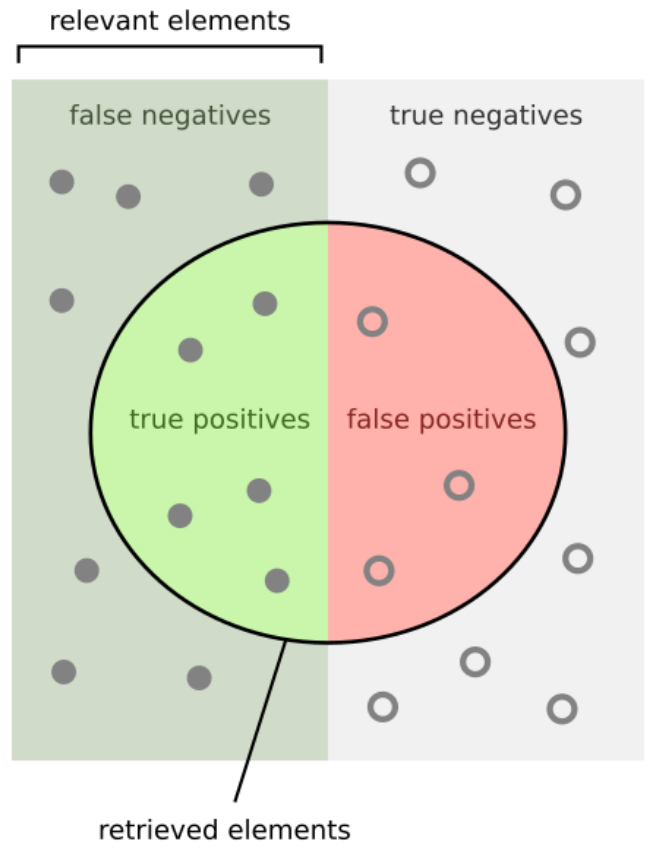


Global Alliance for Genomics and Health (GA4GH)

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Scientific principles in genomics

- Positive controls, negative controls
- Measure false positives, false negatives and report sensitivity, specificity etc
- Appropriate statistics



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

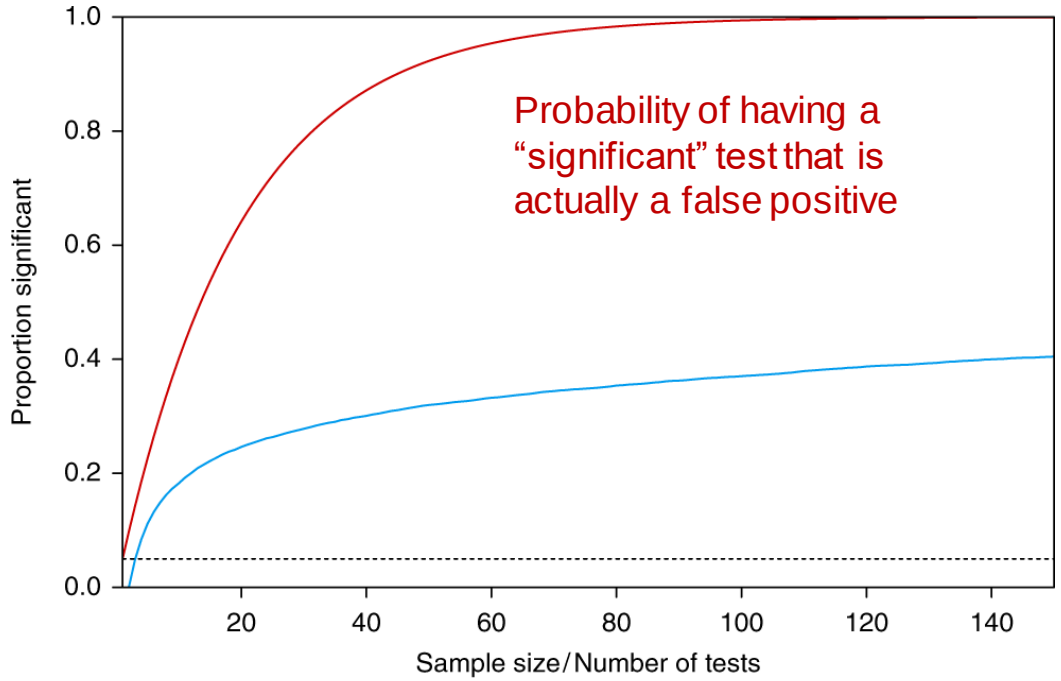
How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

Adapted from Wikipedia.org and manishmazumder5.substack.com

Statistical challenges with large datasets

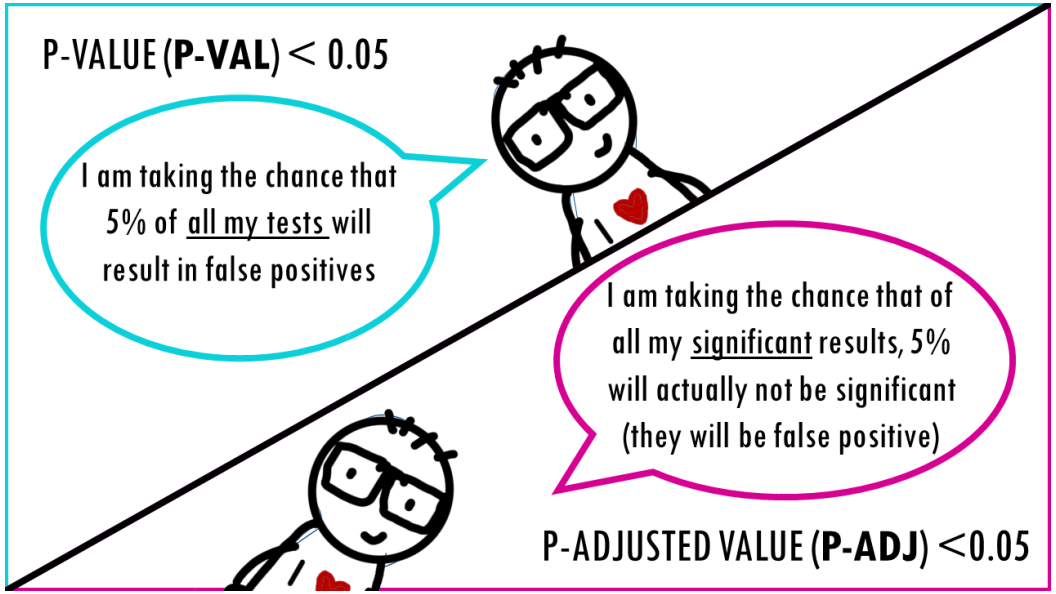
- Statistical significance with alpha = 0.05 assumes you are doing **one test**
- What if you are testing every gene for differential expression? Every organism in a microbiome sample? Every base for the presence of a motif?



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Statistical approaches with large datasets

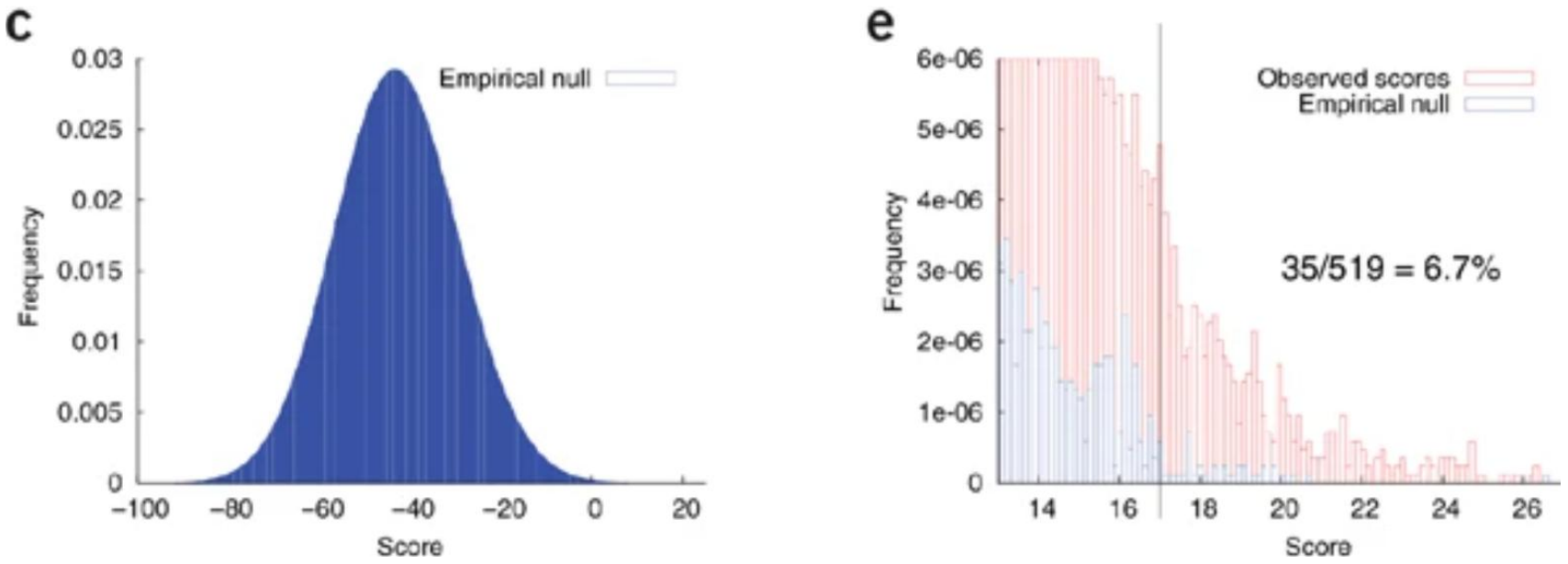
- Consider **Bonferroni correction** or **False Discovery Rate (FDR)** adjusted p value
- A useful collection of commentaries: Nature Methods' Points of Significance https://www.nature.com/collections/qghhqm/pointsofsignificance



https://biostatsquid.com/multiple-testing-correction-fdr/

Statistical approaches with large datasets

- Simulations/bootstrapping/shuffling are useful to make null distributions if your data doesn't follow a known distribution

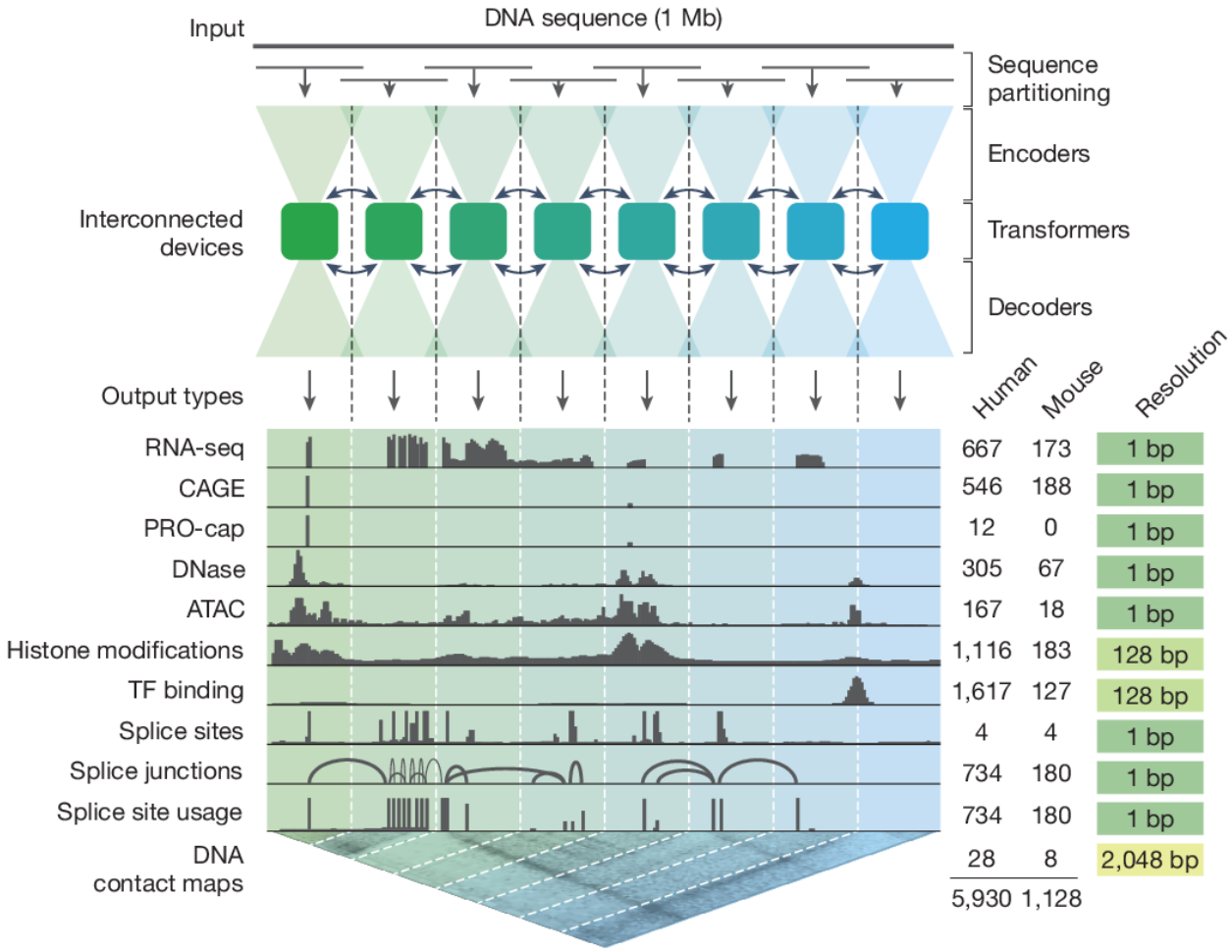


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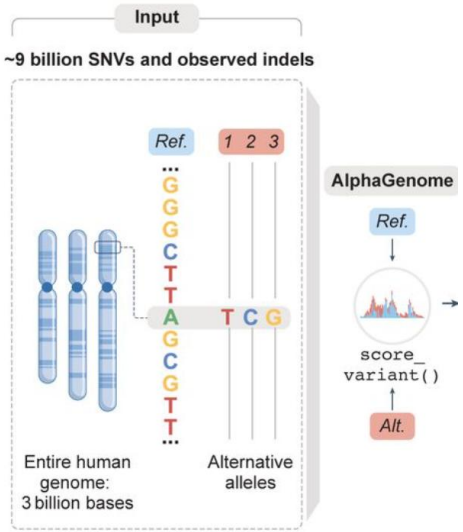
Using AI in genomics

- ML (machine learning) models have been used for decades
  - Things to be aware of:
    - How many features do you have compared to your training data? Do you need to do feature reduction?
    - Is your data overfit?
    - Is the model interpretable? (Eg. use SHAP)
    - Do you have an independent validation set?
- LLMs (large language models) are now being applied everywhere
  - Things to be aware of:
    - Hallucinations
    - Reproducibility
    - Transparency / explainability
    - Be aware of token usage, it can get expensive
    - Privacy

Using AI in genomics



- AlphaGenome recently published predictions for every possible base change in the human genome



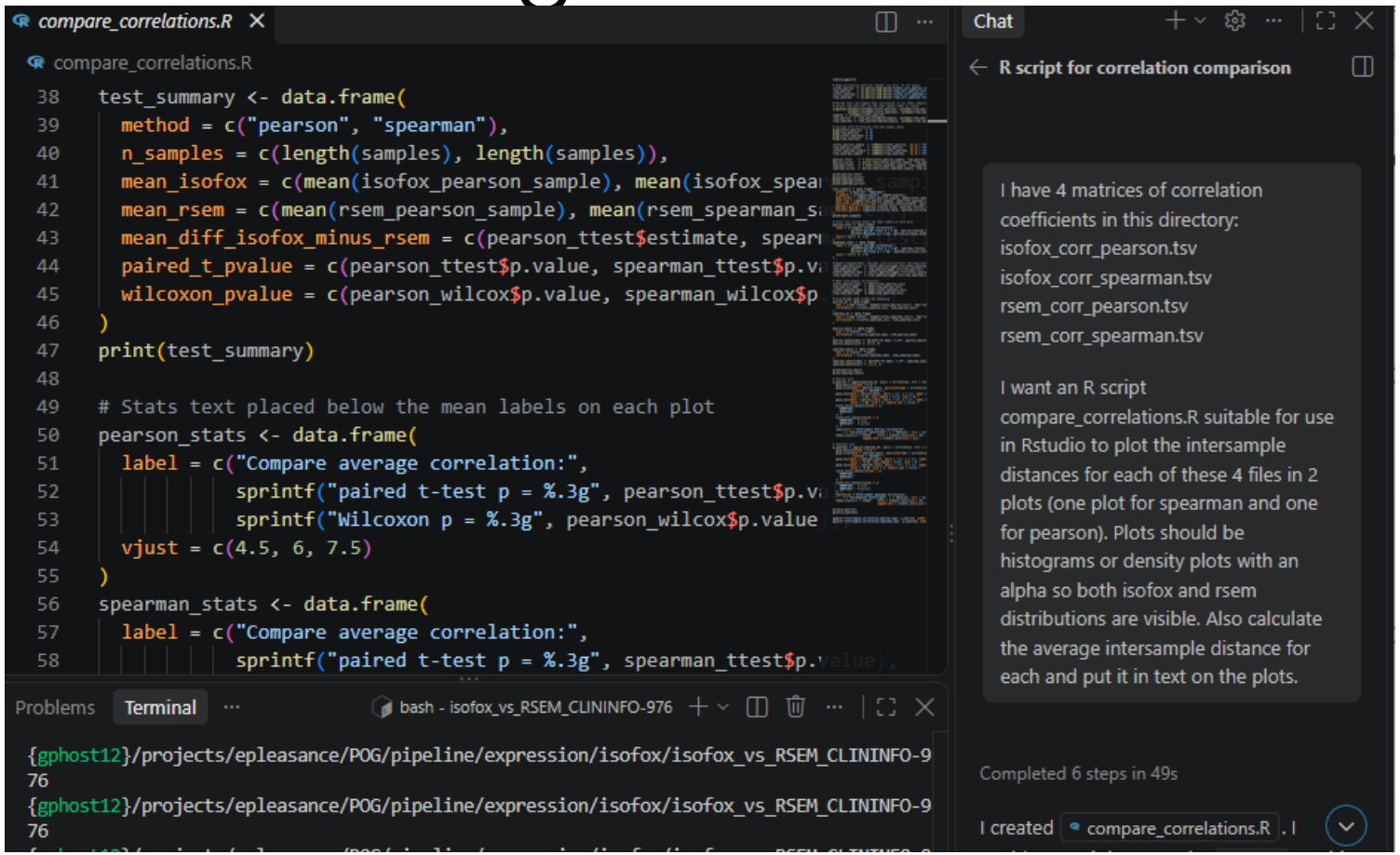
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https://www.medrxiv.org/content/10.64898/2026.09.16.26363192v1

Using AI in genomics: coding

- Code completion
- Documentation
- Code review
- Coding English first (vibe coding)

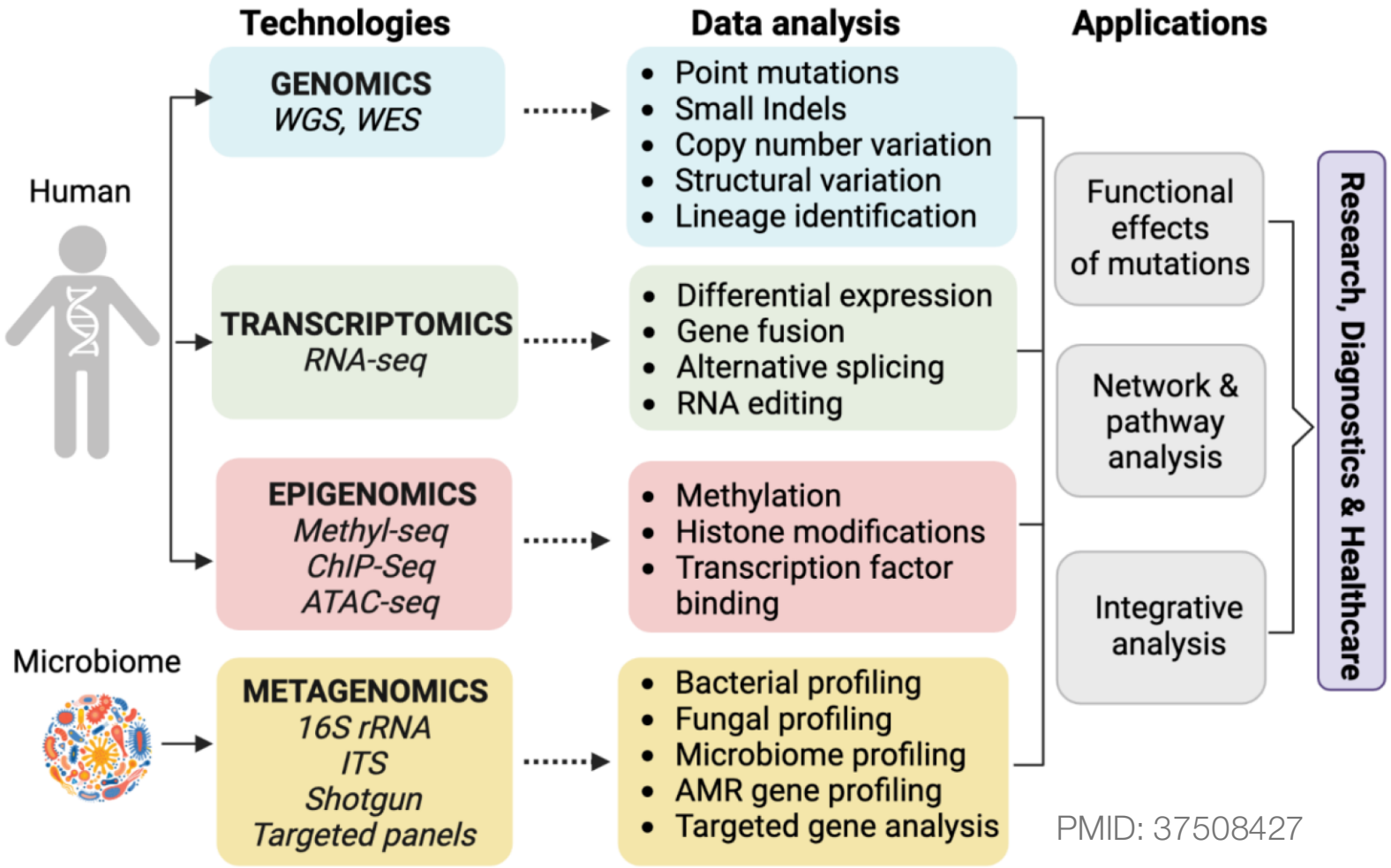
- Examples:
  - Github Copilot
  - Claude Code CLI



- You are ultimately responsible for making sure your code is correct. AI can make assumptions

Sequencing applications

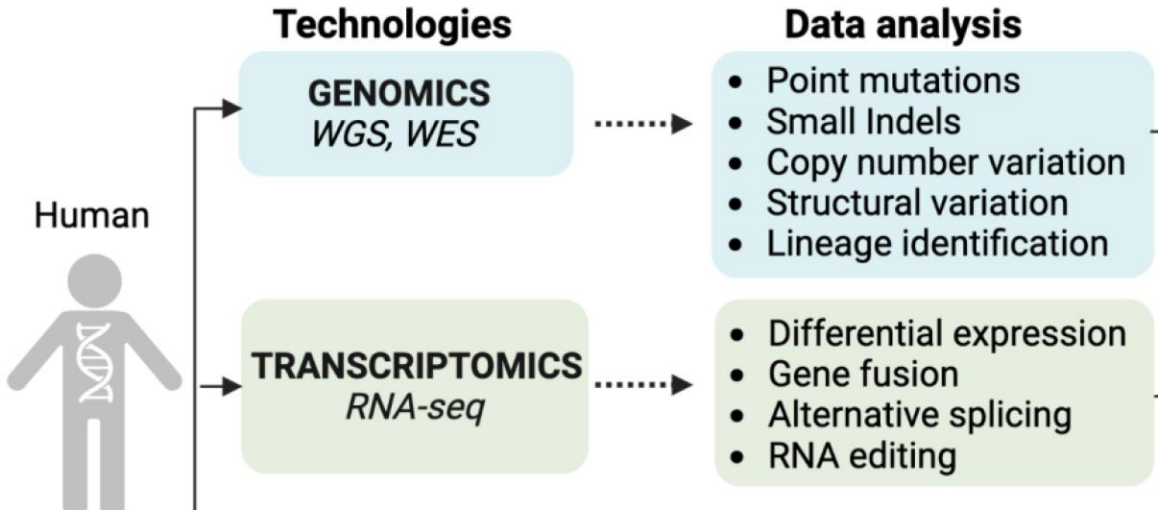
- Type of organism/sample
- Type of molecule
- Specialized prep



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Human genomics

- **WGS**: whole genome sequencing (~3 Gb)
  - Often 30X coverage, more for tumour samples
  - Also commonly used for **novel genome assembly**
- **WES**: whole exome sequencing (~30 Mb)
  - Usually go for 200+ X coverage
- **Targeted panels** (~1-5 Mb; dozens to 100s of genes)
  - Coverage typically > 1000X
  - In library prep or using adaptive sampling
- **RNA-Seq**
  - Usually convert RNA -> cDNA for sequencing
  - Some tech can do direct RNA sequencing

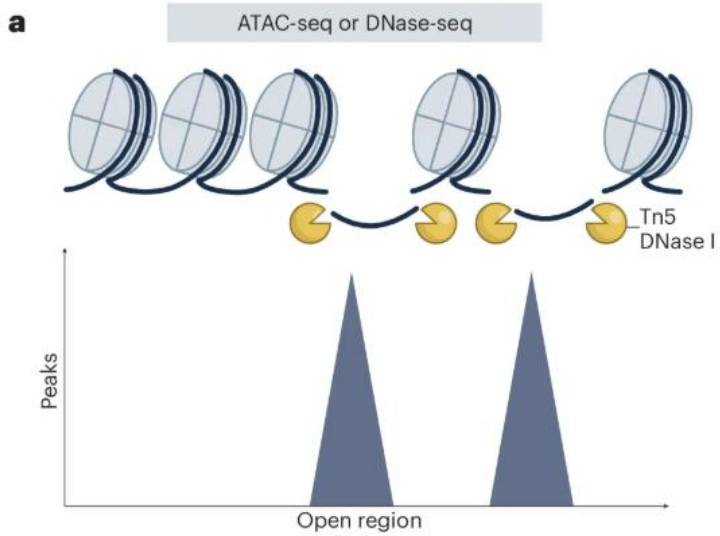


Epigenomics

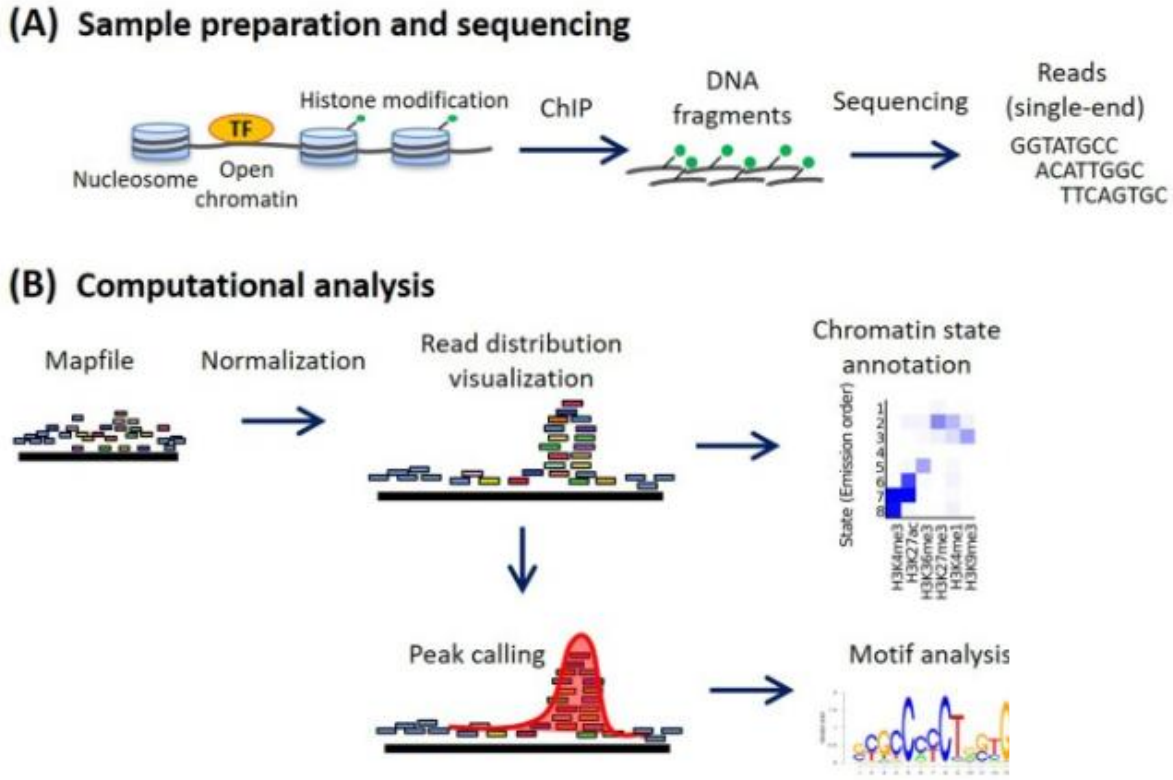
EPIGENOMICS
Methyl-seq
ChIP-Seq
ATAC-seq

- Methylation
- Histone modifications
- Transcription factor binding

- DNA methylation
  - Bisulfite sequencing (Illumina), direct sequencing (Oxford Nanopore, PacBio)
- Histone modifications
  - Chromatin immunoprecipitation (ChIP-Seq)
- Open chromatin
  - ATAC-seq



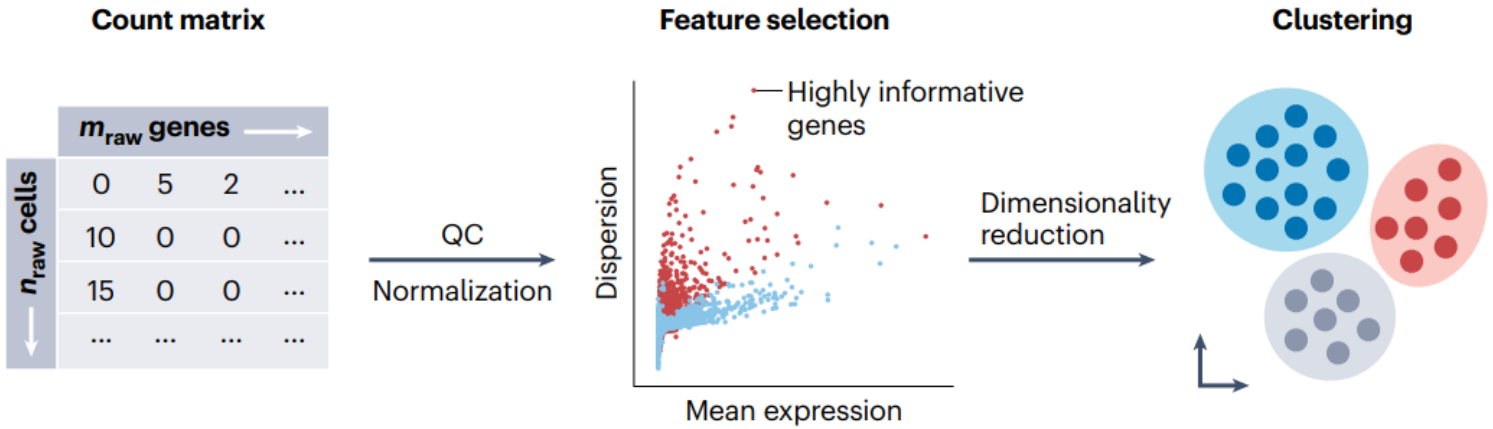
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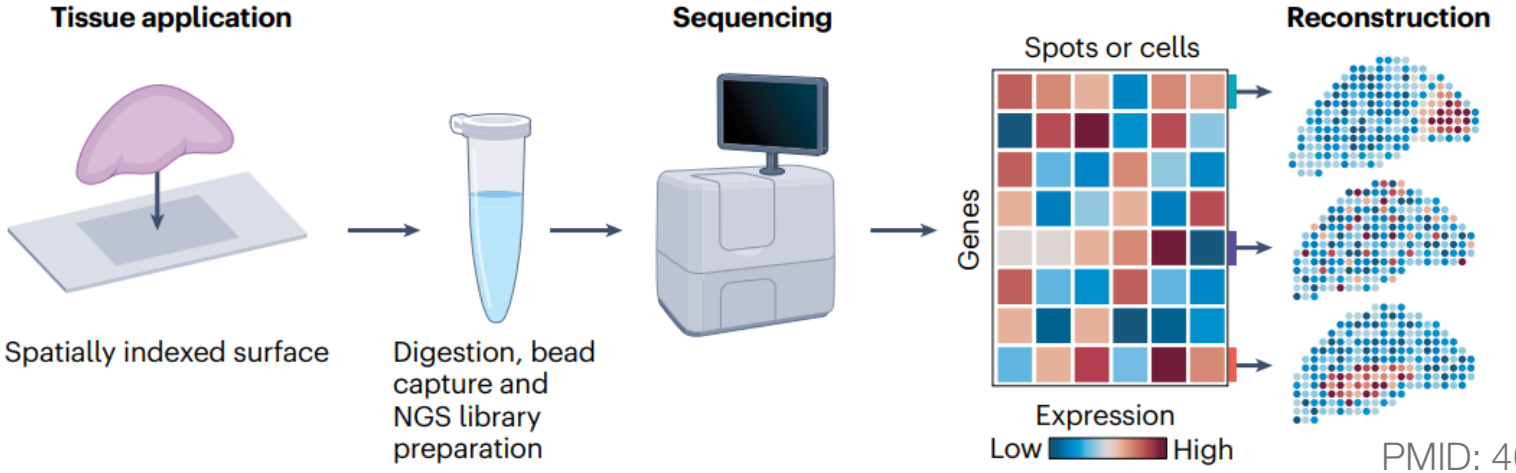
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Specialized approaches

- Single-cell sequencing

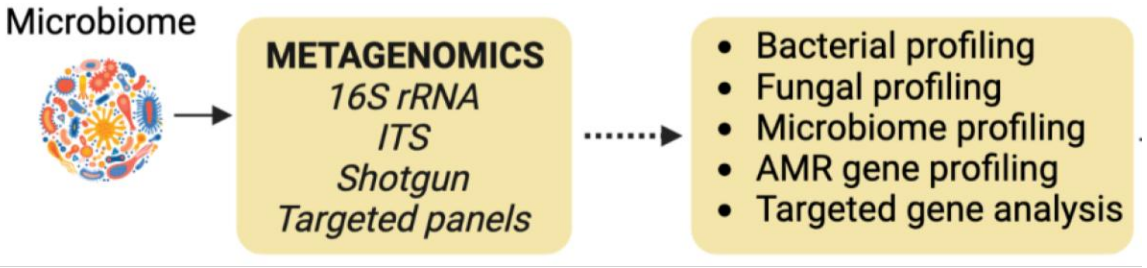


- Spatial profiling

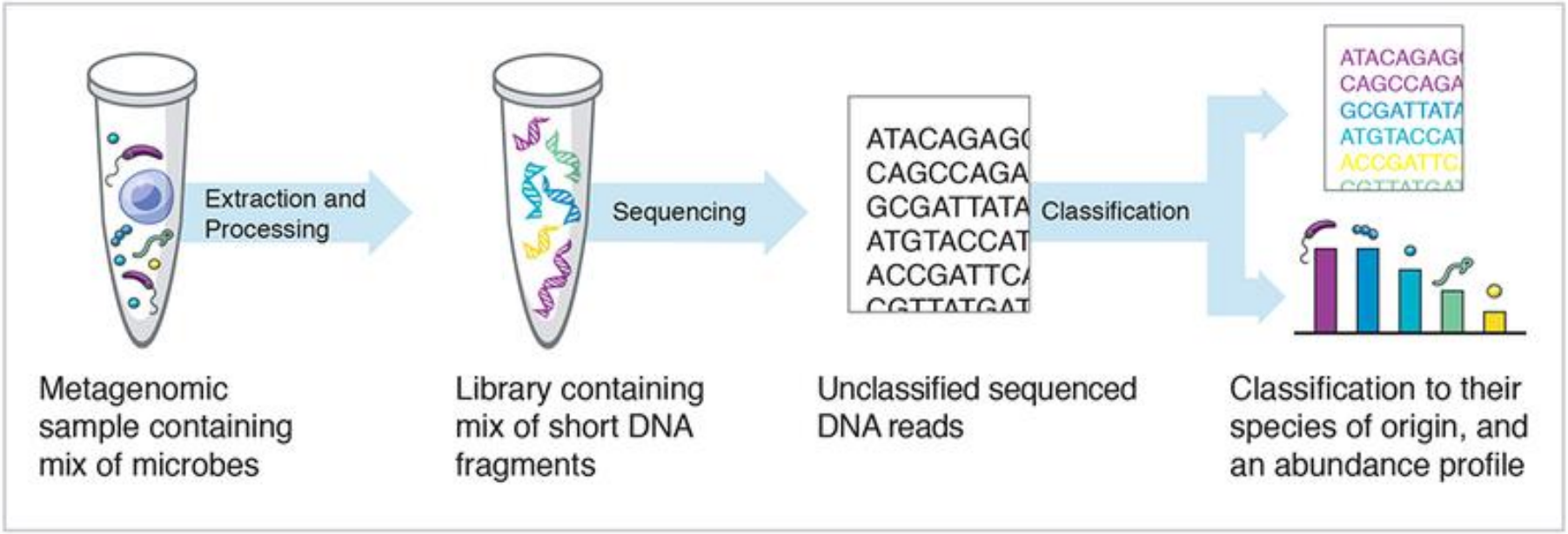


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Metagenomics



- Amplicon-based (16S/18S/ITS)
- Shotgun (short read or long read)



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Scenarios

- We will go through a few example scenarios and discuss what approaches might be relevant
- You will be discussing in groups of 3-4 people
- The slide deck we go through will be posted at the end of today's workshop

Short stretch break

Workshop Sponsors







